

Git

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What is Git?

Version control system (DVCS) used to track changes in source code and manage software development projects.

It was created by Linus Trovalds in 2005 for the development of the Linux kernel.

Simple Definition

Git is a tool that helps developers:

- Track changes in file
- Maintain different versions of a project
- Collaborate with other developers
- Restore previous versions when needed

Why do we need git?

Imagine you are developing a project.

Without git:

```
Project_final.py
Project_Final_new.py
Project_Final_Latest.py
Project_Final_Really_Final.py
```

Managing versions become difficult.

Project=>

```
|=>Git Repository
|=> version 1
|=> version 2
|=> version 3
|=> version 4
```

Git automatically tracks every change.

Features of Git

1. **Version control:** Tracks every modification made to files.
2. **Distributed System:** Every developer has a complete copy of the repository.
3. **Fast performance:** Operations are performed locally.
4. **Branching:** Allows development of new features without affecting the main project.
5. **Collaboration:** Multiple developers can work on the same project simultaneously.
6. **Backup and Recovery:** Previous versions can be restored easily.

Git Architecture

Working Directory

↓

git add

↓

Staging area

↓

git commit

↓

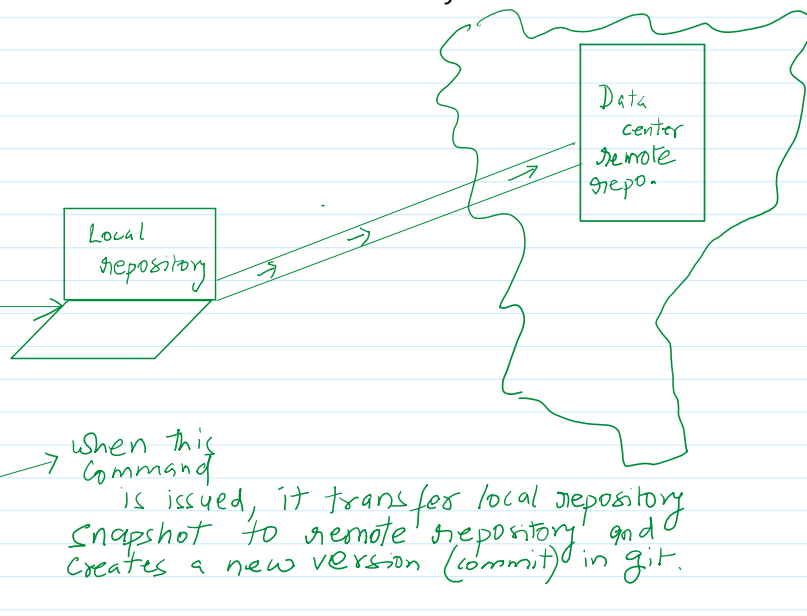
Local repository

↓

git push

↓

Remote repository



Git terminology

Repository (Repo)

A repository is a storage location for your project.

Project files + git History = Repository

Commit: A commit is a snapshot of your project at a specific point in time.

Branch: A branch is an independent line of development.

```
Main
|
|--feature-login
|--feature-payment
```

Merge: Combines changes from different branches.

```
feature-login
|
V
merge
|
V
main
```

WAP to input a number and count number of digits present in it using recursion:

```
def countDigit(num):
    pass
```

#main code

```
N = int(input('Enter any positive integer value:'))
C = countDigit(N)
print(f'Number of digits is: {C}')
```

Output:

Enter any positive integer value: 346

Number of digits is: 3

Handwritten annotations for the recursive function:

- For $num = 99$, the function calls $1 + \text{countDigit}(9)$ (labeled 'main code').
- For $num = 9$, the function calls $1 + \text{countDigit}(0)$.
- For $num = 0$, the function returns 0.

Wap to input number of terms to print and find the sum of following series using recursion:

Number of terms: 3

$$1 + 4 + 9 = 14$$

$$1 + \frac{1}{2} + \frac{1}{3} = 1.08$$

$$1 + 0.5 + 0.3 = 1.08$$

- (a) $1 + 4 + 9 + \dots$
- (b) $1 + \frac{1}{2} + \frac{1}{3} + \dots$
- (c) $1 + \frac{1}{3} + \frac{1}{5} + \dots$
- (d) $\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \dots$
- (e) $2 \cdot 4 + 6 \cdot 8 + \dots$
- (f) $(1^2) + (2^2) + \dots$

